

a4a stochastic slice biomass index

2020-05-13

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1 Data

1.1 Trimmed time series (from 1985)

[...]

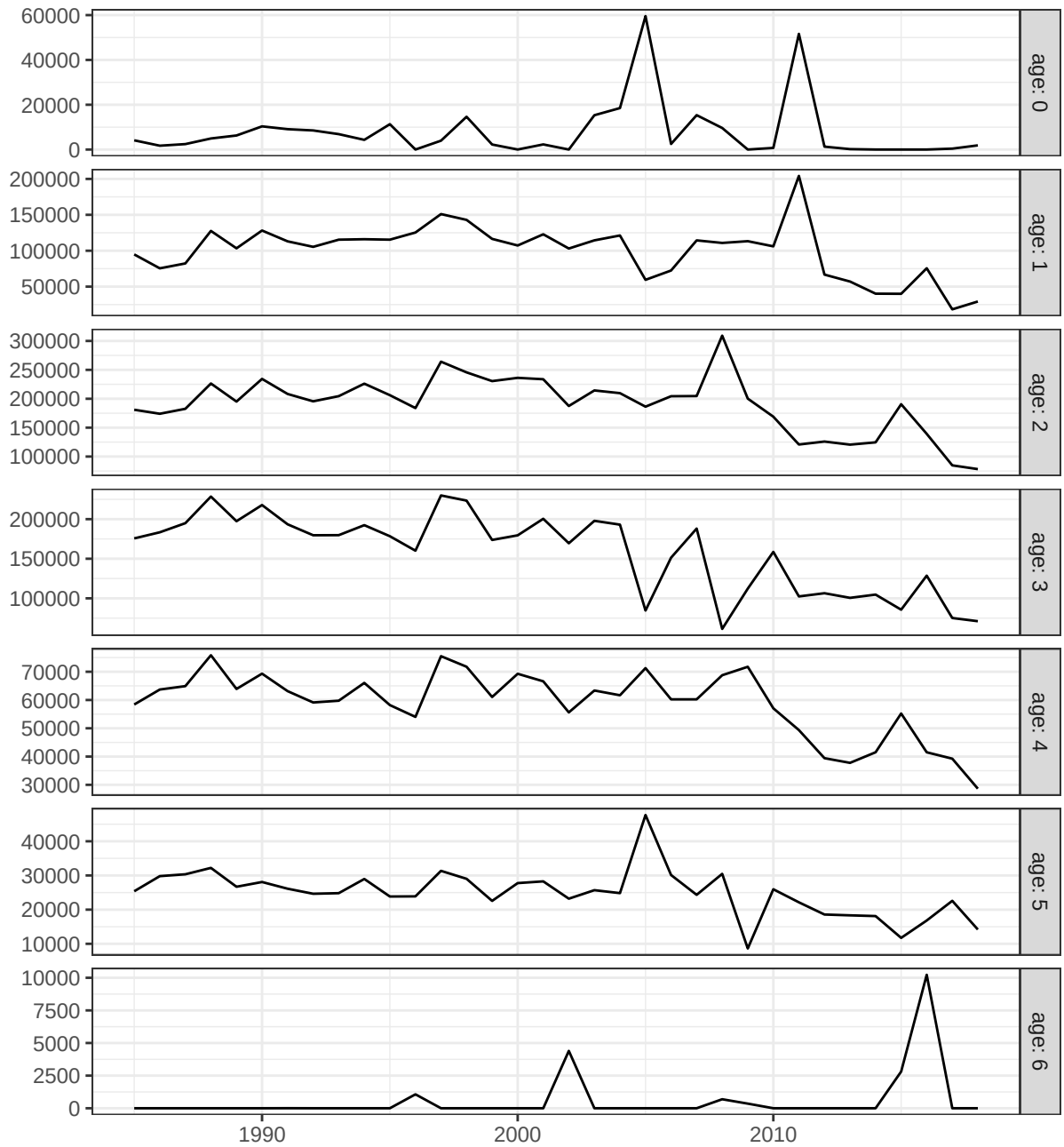


Figure 1. Time series of catch-at-age.

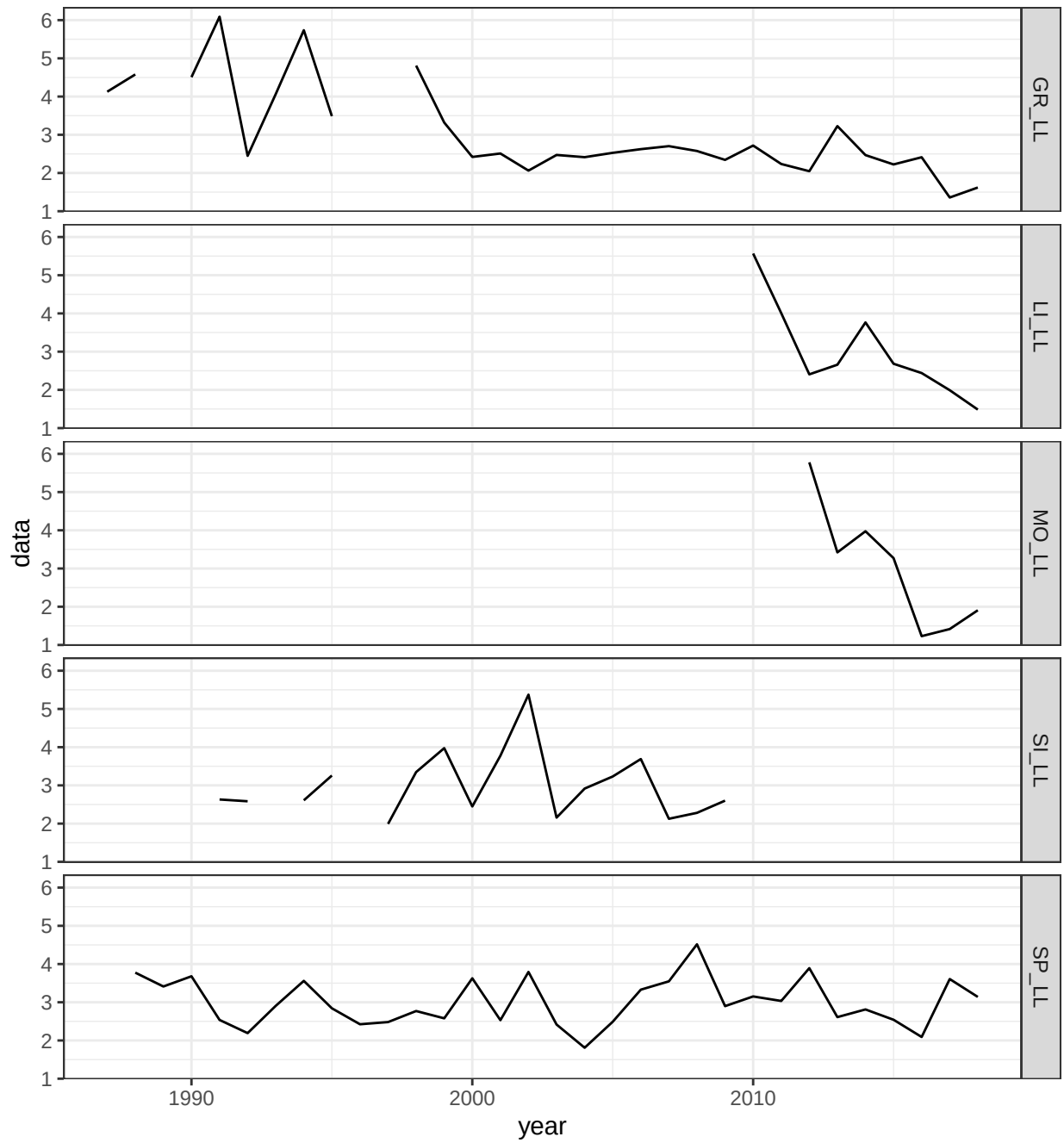
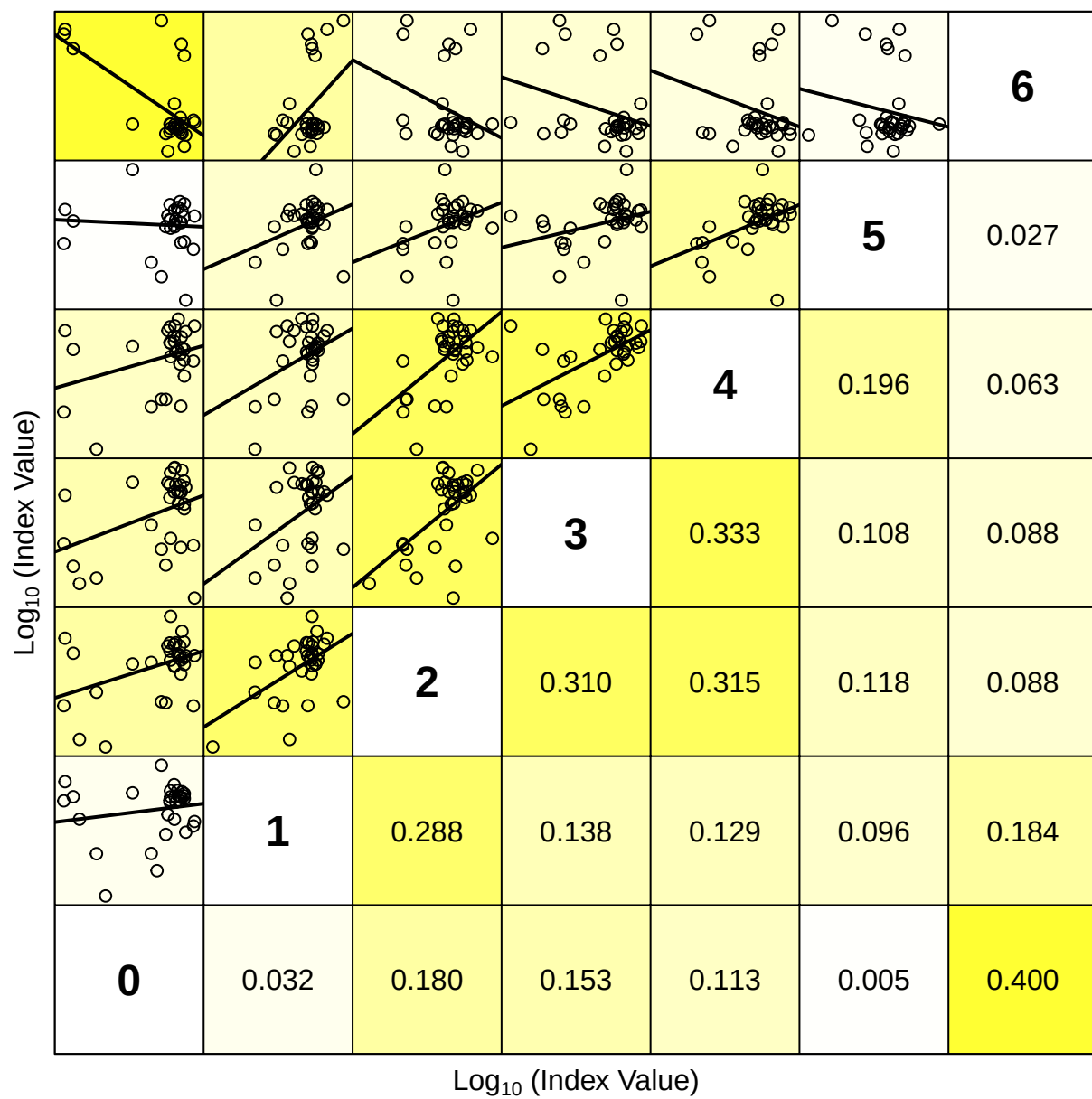


Figure 2. Biomass indices.



Lower right panels show the Coefficient of Determination (r^2)

2 Results fbar 2 - 4

Simple separable factor model on F results in a possible not positive definite covariance matrix which is an indicator of overfitting. The same stands for the built in tensor model. Catchability model is held constant through the years. Absence of SR model gave an unreasonable low recruitment (90.000). Some runs were tested by removing the spanish LLS, the results were almost identical.

2.1 Natural mortality 0.2

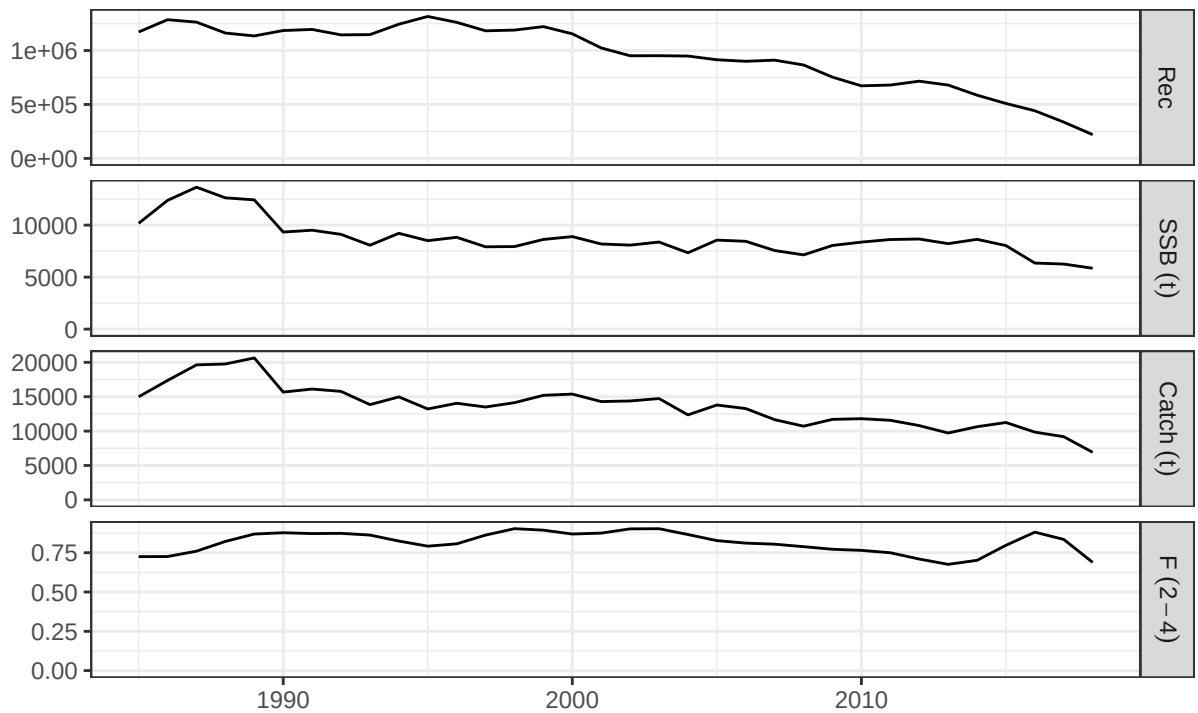


Figure 3. Results for fit 1. F model : $\sim, s(\text{year}, k = 17) + s(\text{age}, k = 3)$, SRmodel: $\sim, s(\text{year}, k = 17)$.

2.2 Lorenzen Natural Mortality

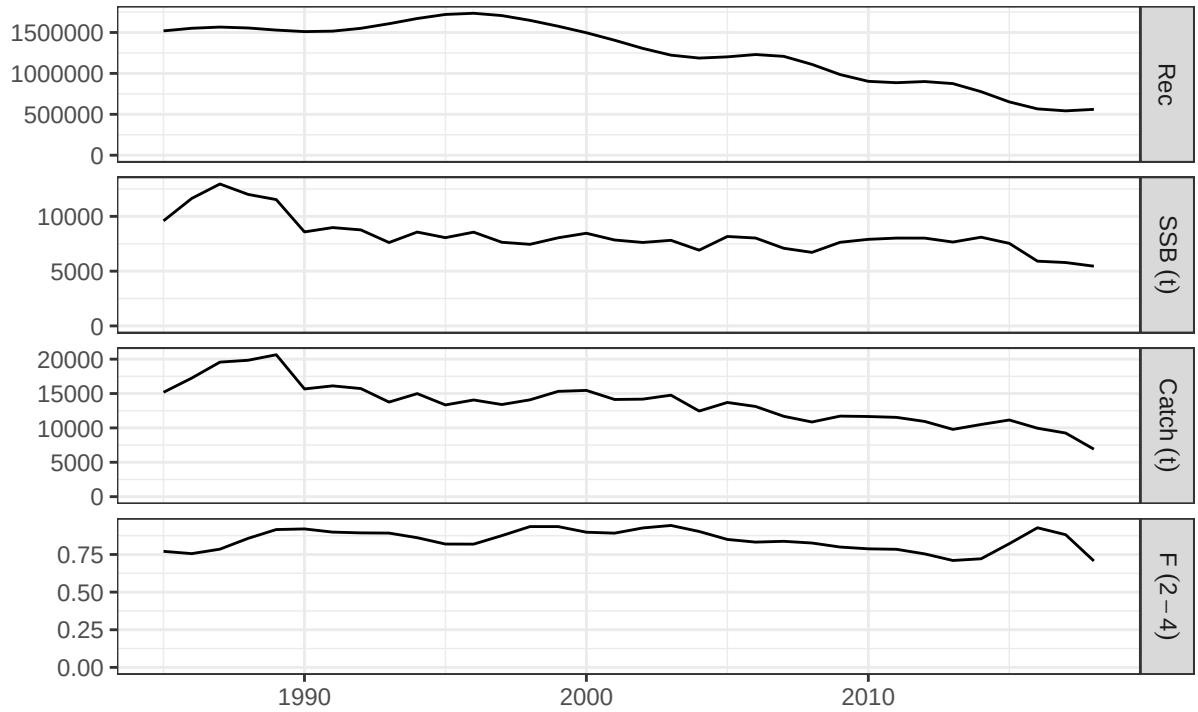


Figure 4. Results for fit 2. F model : $\sim, s(\text{year}, k = 17) + s(\text{age}, k = 3)$, SRmodel: $\sim, s(\text{year}, k = 15)$.

and the corresponding fishing mortalities:

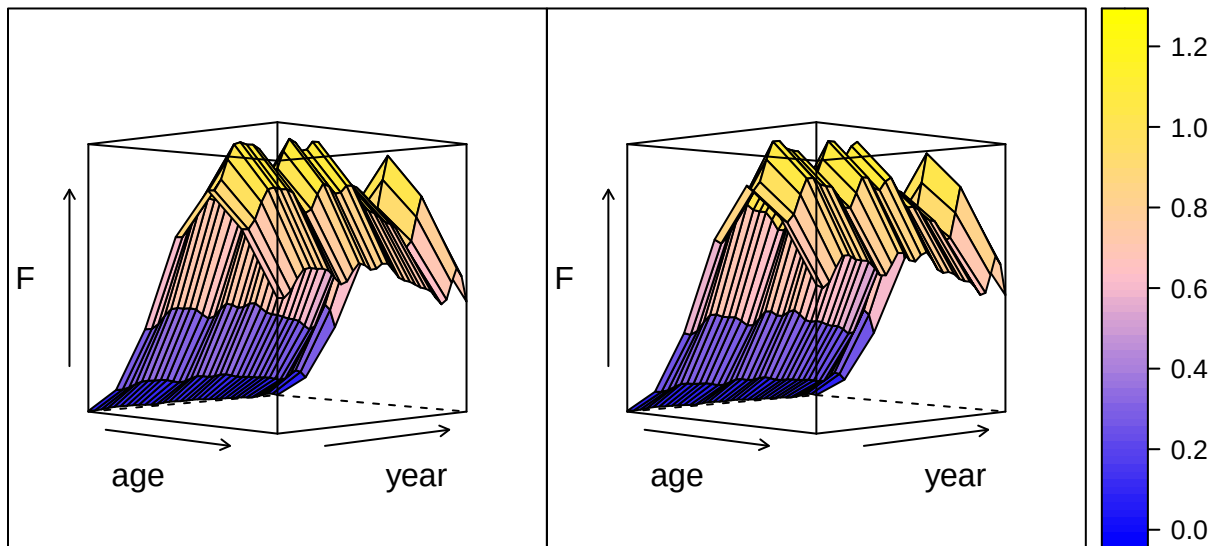


Figure 5. Harvest for $M = 0.2$ (left) and Lorenzen (right).

2.3 Diagnostics (bad fit in catch numbers in age 6)

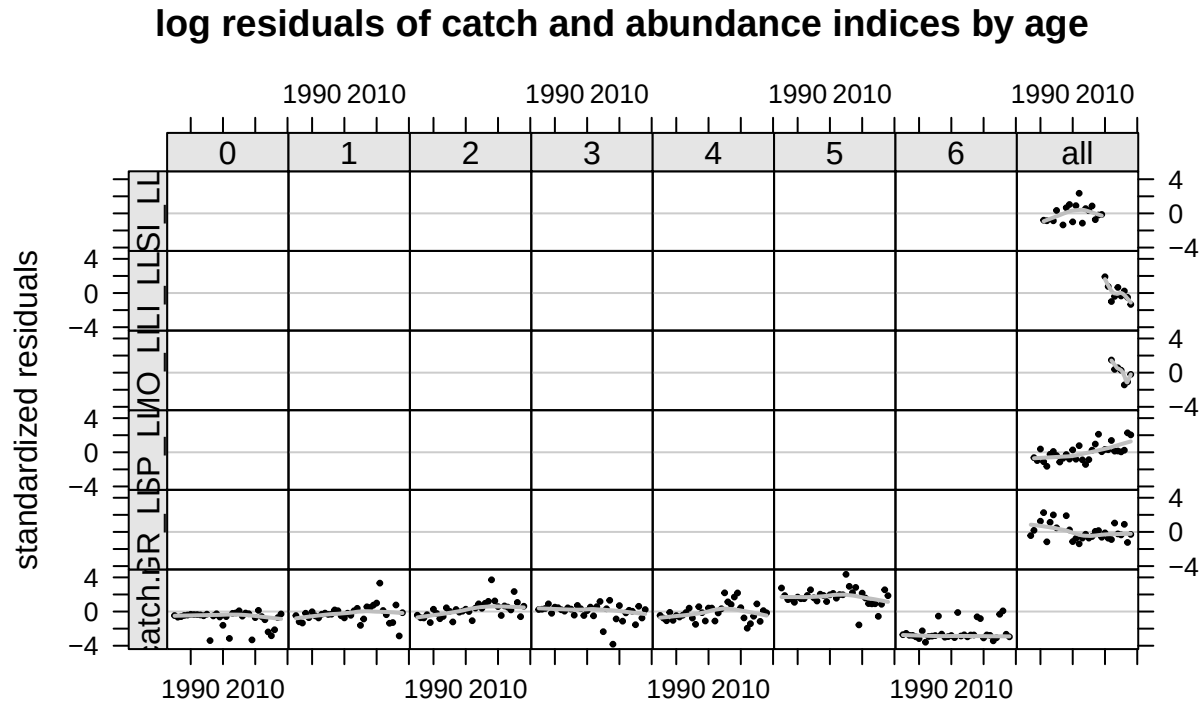


Figure 6. Residuals for fit 1.

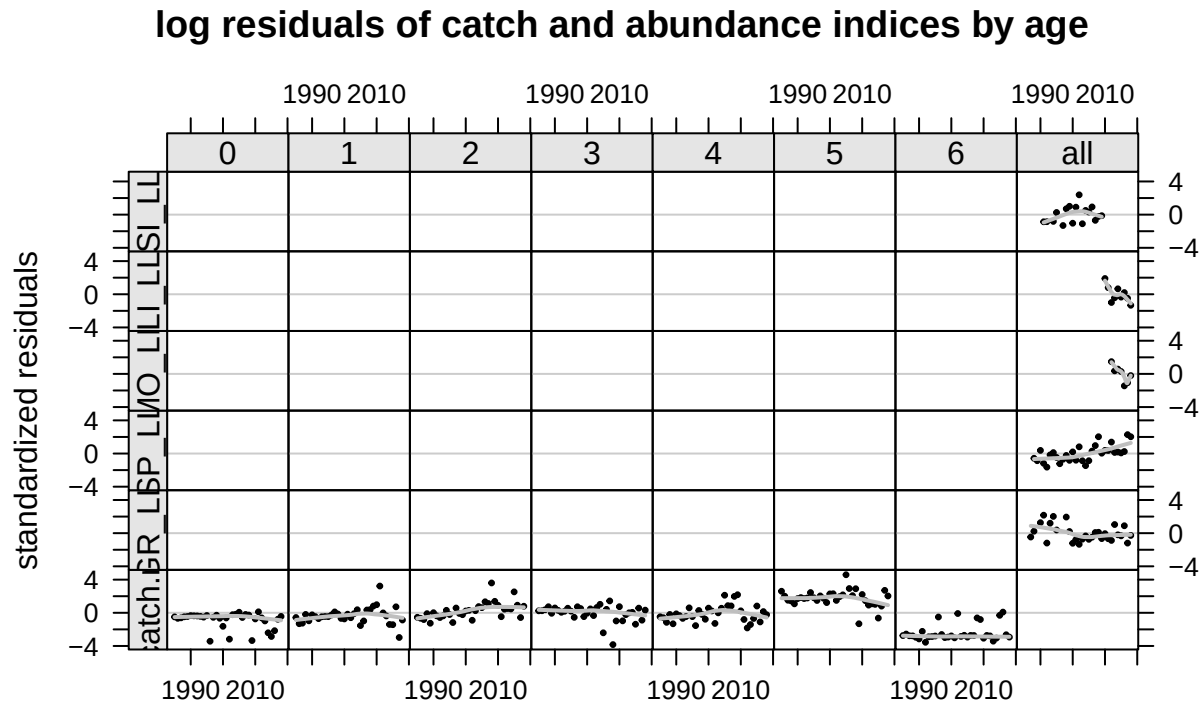


Figure 7. Residuals for fit 2.

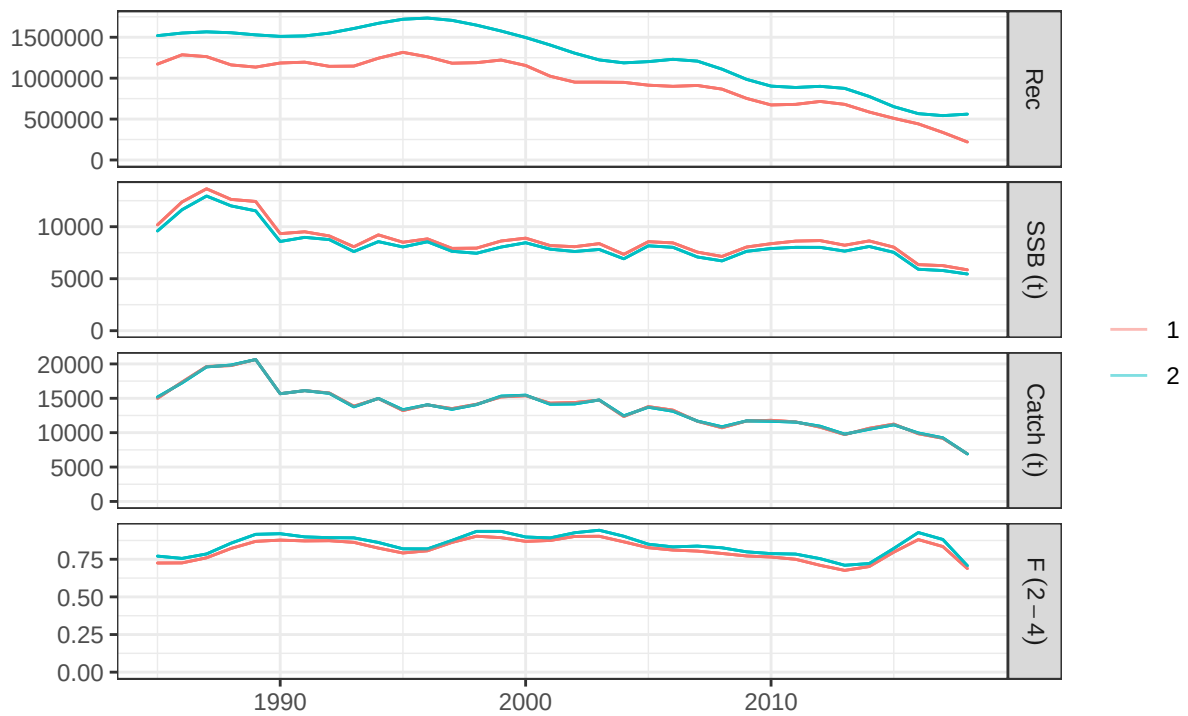


Figure 8. Results for the 2 fits.

```

## Nelder-Mead direct search function minimizer
## function value for initial parameters = -14.516633
## Scaled convergence tolerance is 2.16315e-07
## Stepsize computed as 155424.672844
## BUILD          3 44.464234 -19.904724

## Warning in log(x@.Data): NaNs produced

## HI-REDUCTION    5 35.867465 -19.904724

## Warning in log(x@.Data): NaNs produced

## HI-REDUCTION    7 26.087973 -19.904724

## Warning in log(x@.Data): NaNs produced

## HI-REDUCTION    9 15.486091 -19.904724

## Warning in log(x@.Data): NaNs produced

## HI-REDUCTION    11 5.128231 -19.904724
## LO-REDUCTION    13 -14.516633 -29.365786
## LO-REDUCTION    15 -19.904724 -30.279853
## HI-REDUCTION    17 -26.901761 -30.279853
## HI-REDUCTION    19 -29.365786 -30.279853
## HI-REDUCTION    21 -29.430083 -30.279853
## HI-REDUCTION    23 -29.974312 -30.279853
## HI-REDUCTION    25 -30.010755 -30.279853
## EXTENSION       27 -30.112166 -30.336138
## EXTENSION       29 -30.279853 -30.770690
## LO-REDUCTION    31 -30.336138 -30.770690
## EXTENSION       33 -30.578058 -31.265881
## EXTENSION       35 -30.770690 -31.724904
## EXTENSION       37 -31.265881 -32.539245
## EXTENSION       39 -31.724904 -33.088143
## EXTENSION       41 -32.539245 -33.876945
## LO-REDUCTION    43 -33.088143 -33.876945
## EXTENSION       45 -33.860311 -34.386062
## EXTENSION       47 -33.876945 -34.596723
## EXTENSION       49 -34.386062 -34.744991
## EXTENSION       51 -34.596723 -34.903761
## LO-REDUCTION    53 -34.744991 -34.903761
## EXTENSION       55 -34.888162 -34.988218
## EXTENSION       57 -34.903761 -35.009616
## EXTENSION       59 -34.988218 -35.045760
## LO-REDUCTION    61 -35.009616 -35.045760
## EXTENSION       63 -35.039017 -35.059572
## EXTENSION       65 -35.045760 -35.062032
## EXTENSION       67 -35.059572 -35.067892
## LO-REDUCTION    69 -35.062032 -35.067892
## EXTENSION       71 -35.067488 -35.070643
## EXTENSION       73 -35.067892 -35.071653
## EXTENSION       75 -35.070643 -35.072617
## EXTENSION       77 -35.071653 -35.073268
## EXTENSION       79 -35.072617 -35.073545
## EXTENSION       81 -35.073268 -35.073855
## EXTENSION       83 -35.073545 -35.073922
## EXTENSION       85 -35.073855 -35.074056

```

```

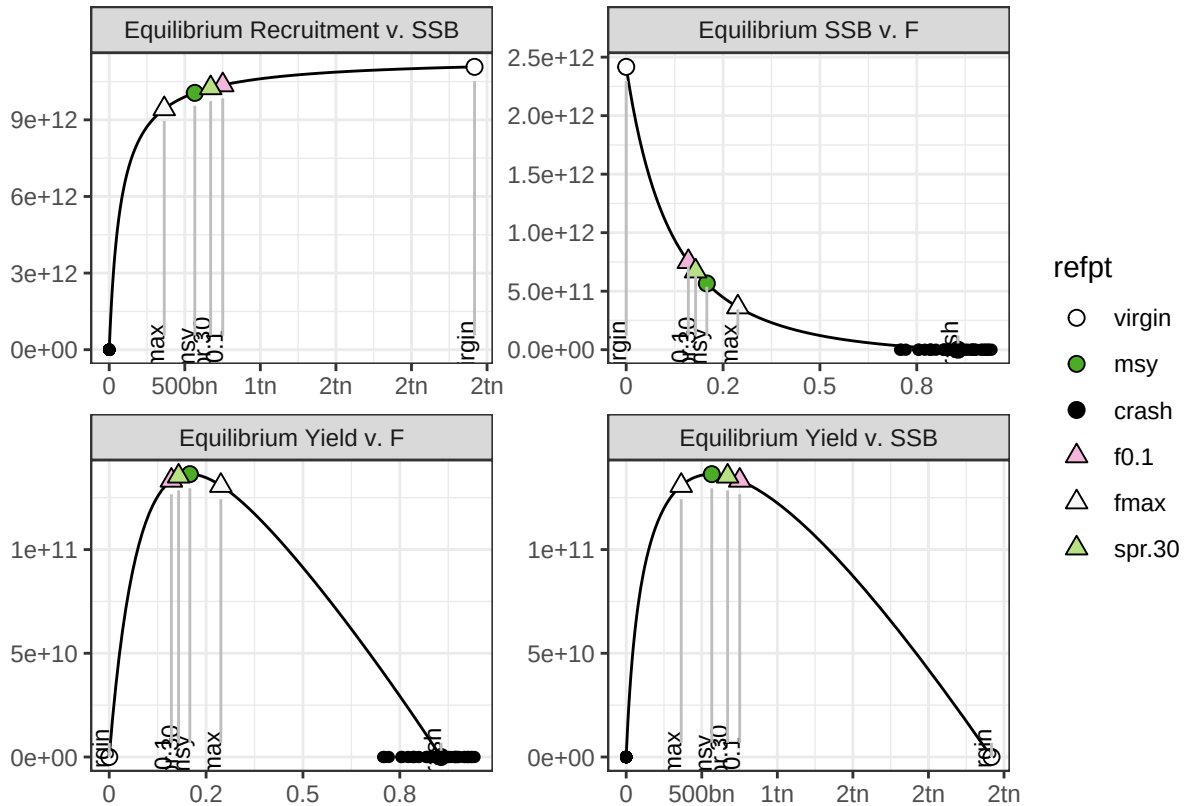
## LO-REDUCTION      87 -35.073922 -35.074056
## EXTENSION         89 -35.074047 -35.074123
## EXTENSION         91 -35.074056 -35.074160
## EXTENSION         93 -35.074123 -35.074180
## EXTENSION         95 -35.074160 -35.074209
## LO-REDUCTION      97 -35.074180 -35.074209
## EXTENSION         99 -35.074206 -35.074227
## EXTENSION        101 -35.074209 -35.074244
## EXTENSION        103 -35.074227 -35.074249
## EXTENSION        105 -35.074244 -35.074266
## LO-REDUCTION      107 -35.074249 -35.074266
## EXTENSION        109 -35.074262 -35.074276
## EXTENSION        111 -35.074266 -35.074292
## LO-REDUCTION      113 -35.074276 -35.074292
## EXTENSION        115 -35.074290 -35.074314
## REFLECTION        117 -35.074292 -35.074325
## REFLECTION        119 -35.074314 -35.074326
## HI-REDUCTION      121 -35.074323 -35.074326
## HI-REDUCTION      123 -35.074325 -35.074326
## EXTENSION        125 -35.074326 -35.074328
## LO-REDUCTION      127 -35.074326 -35.074328
## REFLECTION        129 -35.074328 -35.074328
## HI-REDUCTION      131 -35.074328 -35.074328
## EXTENSION        133 -35.074328 -35.074329
## LO-REDUCTION      135 -35.074328 -35.074329
## EXTENSION        137 -35.074329 -35.074330
## LO-REDUCTION      139 -35.074329 -35.074330
## EXTENSION        141 -35.074330 -35.074330
## EXTENSION        143 -35.074330 -35.074330
## EXTENSION        145 -35.074330 -35.074330
## EXTENSION        147 -35.074330 -35.074331
## Exiting from Nelder Mead minimizer
##      149 function evaluations used

## Warning in if (class(tmphess) == "try-error") {: the condition has length > 1
## and only the first element will be used

## An object of class "FLPar"
##      quant
## refpt  harvest  yield    rec      ssb      biomass  revenue  cost
## virgin 0.00e+00 0.00e+00 1.11e+13 2.42e+12 2.66e+12      NA      NA
## msy    2.08e-01 1.36e+11 1.01e+13 5.66e+11 7.69e+11      NA      NA
## crash  8.57e-01 -2.58e-05 -2.26e-03 -1.53e-05 -5.05e-05      NA      NA
## f0.1    1.60e-01 1.33e+11 1.04e+13 7.51e+11 9.65e+11      NA      NA
## fmax    2.88e-01 1.31e+11 9.43e+12 3.64e+11 5.47e+11      NA      NA
## spr.30 1.79e-01 1.35e+11 1.02e+13 6.71e+11 8.80e+11      NA      NA
## mey      NA      NA      NA      NA      NA      NA      NA
##      quant
## refpt  profit
## virgin      NA
## msy      NA
## crash      NA
## f0.1      NA
## fmax      NA
## spr.30      NA

```

```
##   mey      NA
## units:  NA
## Warning: Removed 1 row(s) containing missing values (geom_path).
## Warning: Removed 2 rows containing missing values (geom_point).
## Warning: Removed 2 rows containing missing values (geom_text).
## Warning: Removed 2 rows containing missing values (geom_segment).
```



```
## Warning in optim(par = c(a = 15542.467284353, b = 74.4822698269999), fn =
## function (par) : unknown names in control: silent
## Warning in log(x@.Data): NaNs produced
## Warning in log(x@.Data): NaNs produced
## Warning in log(x@.Data): NaNs produced
## Warning in log(x@.Data): NaNs produced
## Warning in if (class(tmphess) == "try-error") {: the condition has length > 1
## and only the first element will be used
```

